
MECH 5200 NUMERICAL METHODS FOR PARTIAL DIFFERENTIAL EQUATIONS

NUMERICAL METHODS FOR SOLVING REAL WORLD ENGINEERING PROBLEMS

LECTURE:	Wednesday: 6:30PM - 9:20PM, DAN-220
INSTRUCTOR :	Dr. Valentin Boutrouche
OFFICE HOURS:	On demand
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1 Course Catalogue Description

Mathematical approaches for numerically solving partial differential equations. The focus will be (a) iterative solution methods for linear and non-linear equations, (b) spatial discretization and meshing (c) finite difference methods (FDM), (d) finite volume methods (FVM), (e) finite element methods (FEM) and (f) boundary element methods (BEM). The theory behind of each of these methods will be developed and discussed. Computer programming applications involving the solution of linear and non-linear PDEs in multiple dimensions will play a key role in this course. Unique computer programming assignments will be selected from different engineering/science fields (possibilities include: fluid flow, heat transfer, electrostatics, electromagnetism, structural analysis, medical, ocean engineering etc.) to illustrate the broad applicability of numerical methods. Students will be expected to complete programming assignments – while most class examples will deal with pseudo code and/or MATLAB, a working knowledge of one of the following programming languages is recommended: MATLAB, Octave, C, C++, fortran, Java, BASIC, or Python.

2 Course Summary

This course is a combination of *mathematics* (partial differential equations, numerical methods), *engineering* (engineering relevant PDEs and applications) and *computer science* (development and coding of algorithms for solving PDEs using computers).

By the end of this course, you should be able to use the main numerical/computational approaches to solve multi-dimensional partial differential equations using a computer.

3 Assessment

- 20% : In-class exercises
- 60% : Homework projects w/ code
- 20% : PDE solver project (choice).

4 Descriptions of Assignments

- **In-class exercises:** In-class exercises will be performed: on paper, in Blackboard/ Gradescope, and using the MATLAB grader embedded in Blackboard. These exercises will be both individual and group-work.
- **Homework projects w/ code:** The homework for this class will comprise multiple (4-6) significant homework projects. You should plan ahead and make consistent progress on each homework project once it is assigned – *they are not designed to be done at the last minute.*
- **PDE solver project (choice):** Students may work as individuals or in groups up to 4 people on a final PDE course coding and writing project. You will select a topic area, research it, develop a strategy for numerical solution of the PDE, and then implement the solution. Each group/individual will submit a typed technical report similar to a short conference paper outlining their project.

This course does not have a final exam nor does it have in-class tests. Despite this, the course is challenging due to the content and implementation of computer code.

4.1 Assignment Submission and Learning Management System

- **Blackboard:** The course will use Blackboard as a primary location for course content and assignments. You should familiarize yourself with the Blackboard environment early and seek help as needed. The assigned readings, MATLAB problems, videos, learning checks and links to any projects will be listed in folders in blackboard.
- **Gradescope:** Gradescope will be used for the submission of your written project components throughout the semester. If you have not used Gradescope before, please familiarize yourself with the Gradescope interface and submission of assignments online.

Gradescope student center:

<https://help.gradescope.com/category/cyk4ij2dwi-student-workflow>

Short tutorial:

https://www.gradescope.com/get_started#student-submission

How to submit PDFs:

https://gradescope-static-assets.s3.amazonaws.com/help/submitting_hw_guide.pdf

- **MATLAB:** You are expected to engage with MATLAB during homework and projects. In addition, you will be provided additional MATLAB codes/apps to learn MATLAB and course content.

- **MATLAB Grader:** We will be using the Blackboard integrated MATLAB Grader this semester to assess/ grade your MATLAB code. You will be given unlimited attempts to submit your scripts/functions for credit. You should make full use of this capability to test and improve your MATLAB submissions.

4.2 Course Deployment

The course will be taught in a **blended learning** format. There are video modules which are assigned as homework and are to be watched **prior to the class** on the topic.

Class time will be used to explore the concepts covered in the videos using worked out programming examples and student driven programming problems. You are encouraged to bring a laptop and note-paper to class so that you can participate in these in-class hands-on activities.

The approximate course topic schedule is:

Week	Topic
1-2	Introduction to MATLAB
1	Course Intro/Algebra/MATLAB/FD & Common PDEs
2	Linear Algebra, Applying 1-D Finite Diff. equations
3	Coding 1D Finite Differences.
4	2D Domain Mappings + Implement Example 2D
5	Explicit and Implicit Parabolic Equations
6-9	Finite Volume Methods
10 - 13	Finite Element Methods
14	Boundary Elements for PDEs and/or DG
15	End of semester expo

4.3 Goals for the course

The course is divided into several sections/categories of material, which includes: Mathematical methods, linear algebra and solutions of systems of equations; Geometry definition and discretization of PDEs; Finite Difference Methods; Finite Volume Methods; Finite Element Methods; Boundary Element Methods (time permitting). Section content will be evaluated using homework assignments. Your semester project will also cover these areas in more depth.

5 Textbook and Reference Material

5.1 MATLAB or similar programming tools

The university provides all enrolled students with access to an annual (1-year) license to MATLAB. To download and install MATLAB, follow the instructions found here:

<https://www.uml.edu/it/software/>

Goal	Course Outcome	ABET Criterion
1	Understand partial differential equations (PDEs) and their application to practical engineering phenomena, such as: heat transfer, fluid flow, electrochemistry, structural mechanics.	1
2	Apply different numerical methods to solving PDEs, including the Finite Difference, Finite Volume, Finite Element, and Boundary Element Methods.	1,7
3	Code numerical methods in a programming language, such as MATLAB or Python.	1,7
4	Study effectively and efficiently in groups to perform the group project.	5
5	Understand how to extend programs from 1D models to Multiple-dimensions.	1,6,7
6	Understand the similarities vs. differences, advantages vs. disadvantages among different numerical methods and their implementations.	6,7

Some students may wish to explore other programming languages. One popular language which is a free, but quite popular alternative to MATLAB is *Python* with appropriate libraries added.

If you choose to use Python, the faculty support will be considerably less. However, if you know MATLAB really well, then Python represents a good learning opportunity/alternative programming language.

5.2 Course Reference Books

There is no textbook required for this course. The following books are recommended for FDM and FEM methods (you can likely find copies in the library):

- J. Li and Y-T Chen, *Computational Partial Differential Equations Using MATLAB*, Chapman and Hall/CRC press, ISBN: 978-1-4200-8904-2.
- Tannehill, Anderson, and Pletcher, *Computational Fluid MEchanics and Heat Transfer*, Taylor & Francis, 1997.

The following textbooks are recommended for Finite Volume Methods:

- R.J. Leveque, *Numerical methods for conservation laws*.
- H. Versteeg and W. Malalasekera, *An Introduction to Computational Fluid Dynamics: The Finite Volume Method (2nd Edition)*.

A highly recommended and referenced source for additional lecture notes and materials is the 16.901/16.910/16.920 course lecture note sets on MIT's open course-ware (OCW). **The notes from MIT's course 16.920 are referenced and are a source for deeper learning and engagement with the material**

<https://ocw.mit.edu/courses/16-920j-numerical-methods-for-partial-differential-equations-sma-5212-spring-2003/pages/lecture-notes/>

<https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-336j-introduction-to-numerical-simulation-sma-5211-fall-2003/>

5.3 Computer Programming Project

You will perform a semester project that involves proposing and solving a PDE of your choice using 1-2 of the three methods FDM, FVM or FEM. The project will comprise:

1. **Project proposal (10%)**: You could choose to work individually or as a group with up to 4 other students. Up to four students form a group. Each group/individual must submit a first project proposal by Friday Feb. 7th. Depending on the quality and subject of the proposal, you may be asked to submit a revised proposal.
2. **Computer code (20%)**: You must submit working (MATLAB or similar) computer codes that address the proposed problem.
3. **Final technical report (60%)**: You must submit a typed technical report including the technical explanation of: (1) The physics of your problem, (2) The mathematics/numerical methods used, (3) The software/code designed and (4) validation of the results.
4. **Presentation (10%)**: Slides must be created on the basis of the final report and briefly presented in class.

5.4 Homework & Homework Policy

The homework will cover the main concepts discussed in the videos and in class. Homework will explore both theory and application of the numerical methods learned and will culminate in the development and use of computer code. All homework **will have a written and coded component** (to be handed in pdf) and some will have a coding component (to be automatically graded in the MATLAB grader). **Homework assignments must be neatly written or typed.**

While group interaction in the form of discussions and study-groups is highly encouraged, the homework that is handed-in **must** be performed individually. This means that both the computer code and the written questions must be individual efforts. *Since the homework is the source of the majority of your grade, a zero-tolerance policy will be in effect in accordance with the UML academic integrity policy:*

<https://www.uml.edu/Catalog/Undergraduate/Policies/Academic-Policies/Academic-Integrity.aspx>

5.5 Attendance and Participation:

Students are expected to attend class regularly, as regular attendance is one of the most important contributors to student success. However, students may occasionally need to miss class due to illness, emergency, or caring for a sick family member. Do not come to class if

you are ill. I try to be very accommodating to students who are experiencing attendance challenges, but you must communicate your situation with me regularly and with as much advance notice as possible.

Likewise, if I should need to miss class, I will communicate with you via Email as soon as possible with clear instructions.

If the campus is closed due to weather or other unforeseen circumstances, please check your email for information on how to proceed. Class may be cancelled, held virtually, or course material may be presented in a different way in order to stay on schedule.

5.6 Student Mental Health and Well-being

Your personal health and well-being are important to all of us at the university. I'm available to talk about your stresses or concerns related to your coursework in my class.

Here are some resources to support your well-being:

Counseling Services provide crisis intervention, assessment, referrals, short term individual counseling and group therapy. Call to book an appointment at (978) 934-6800.

UMatter2 is a university-wide initiative to support students and promote mental health. They can be reached at (978) 934-6671. There, you also will find information at their website on how to access Togetherall, an online community which is a peer-to-peer platform dedicated to mental health support.

Centers for Learning and Academic Support Services (CLASS) provides advising services including goal setting, course selection, SIS functions, changing majors/minors and course deletions. (978) 934-2936 or

Advisement@uml.edu

The mission of the Office of Student Life & Wellbeing is to advance the holistic concept for student success by infusing health-promoting actions and collaboration into campus culture. They can be reached at 978-934-4342 or:

Wellbeing@uml.edu

5.7 Disability Services

If you are registered with Disability Services and will require course accommodations, please notify me via the Accommodate semester request process as soon as possible so that we might make appropriate arrangements. It is important that we connect to discuss the logistics of your accommodations; please speak to me during office hours or privately after class as I respect and want to protect your privacy. If you need further information or need to register for academic accommodations, please visit the Disability Services Website.

5.8 Diversity, Inclusion, and Classroom Community Standards:

UMass Lowell—and your professor—value human diversity in all its forms, whether expressed through race and ethnicity, culture, political and social views, religious and spiritual beliefs, language and geographic characteristics, gender, gender identities and sexual orientations, learning and physical abilities, age, parenting status and social or economic backgrounds. Enrich yourself by practicing respect in your interactions, and enrich one another by expressing your point of view, knowing that diversity and individual differences are respected, appreciated, and recognized as a source of strength.

The Office of Multicultural Affairs (OMA) supports and advocates for students while leading diversity-related programming. At the same time working to create an inclusive environment for LGBTQ+ individuals via the LGBTQ+ Resource Center. Contact (978) 934-4336 or:

`Multicultural_Affairs@uml.edu`.

5.9 Academic Integrity Policy:

All students are advised that there is a University policy regarding academic integrity. Students are responsible for the honest completion and representation of their work. See the Homework Policy for specific details pertainign to this course.

UMass Lowell's academic integrity policy is:

<https://www.uml.edu/catalog/graduate/policies/academic-integrity.aspx>

5.10 Cell Phones and Other Devices:

You may use your cellphone and other devices (e.g., laptops) during class to look-up class relevant information and submit check-ins. If you need to take (urgent/job related) phone calls during class time, we ask that you respect your peers' learning environment and quietly exit the class to take the call.

5.11 University Privacy Statement

UMass Lowell recognizes the importance of mutual trust between students and faculty. Massachusetts is a two-party consent state, which means it is illegal to record someone without their permission. Recordings of classroom lectures are the intellectual property of the instructor. Instructors have the right to prohibit audio and video recording of their lectures, unless the requesting student is registered with Disabilities Services and recording of class sessions is an approved accommodation. In addition, sharing or selling recordings of classroom activities, discussions or lectures with any other person or medium without permission of the instructor is prohibited, and may result in sanctions for academic dishonesty or referral to the dean's or provost's office.